250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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Cell2Fire: A Cell-Based Forest Fire Growth Model to Support Strategic Landscape Management Planning

Cristobal Pais · Jaime Carrasco · D. Martell · A. Weintraub · D. L. Woodruff

Environmental Science · Frontiers in Forests and Global Change · 22 May 2019

Cell2Fire is a new cell-based **wildland fire** growth simulator designed to integrate data-driven landscape **management** planning **models**. The **fire** environment is **modeled** by partitioning the landscape into… Expand

47 [PDF] arXiv Save Cite

Local-Scale Fuel-Type Mapping and Fire Behavior Prediction by Employing High-Resolution Satellite Imagery

G. Mallinis · I. Mitsopoulos · A. Dimitrakopoulos · I. Gitas · M. Karteris

Environmental Science · IEEE Journal of Selected Topics in Applied Earth… · 1 December 2008

TLDR This study presents an integrated approach to **forest fire management**, combining very high spatial resolution imagery with **fire behavior simulation**, **and** develops a detail site-specific fuel **model** in a Mediterranean area that is suitable **for fire behavior prediction**. Expand

34 IEEE Save Cite

Review and perspectives of digital twin systems for wildland fire management

Yizhou Li · Tianhang Zhang · Yifei Ding · Rahul Wadhwani · Xinyan Huang

Environmental Science · Journal of **Forest** Research · 14 December 2024

Effective **wildland fire management** requires real-time access to comprehensive and distilled information from different data sources. The Digital Twin technology becomes a promising tool in… Expand

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Advancements in Artificial Intelligence Applications for Forest Fire Prediction

Hui Liu Lifu Shu Xiaodong Liu Pengle Cheng Mingyu Wang Ying Huang

Environmental Science, Computer Science · Forests · 19 April 2025

TLDR This study aims to provide valuable support for forest natural resource management and enhanced environmental protection through the application of remote sensing technologies and artificial intelligence algorithms. Expand

3 Publisher Save Cite

Research Trends in Wildland Fire Prediction Amidst Climate Change: A Comprehensive Bibliometric Analysis

Mingwei Bao Jiahao Liu +4 authors Jianxiang Liu

Environmental Science · Forests · 10 July 2024

Wildfire prediction plays a vital role in the management and conservation of forest ecosystems. By providing detailed risk assessments, it contributes to the reduction of fire frequency and severity,… Expand

5 Publisher Save Cite

Utilizing MODIS remote sensing and integrated data for forest fire spread modeling in the southwest region of Canada

H. Dastour Quazi K. Hassan Environmental Science ·

Environmental Research Communications · 31 January 2024

Accurate prediction of fire spread is considered crucial for facilitating effective fire management, enabling proactive planning, and efficient allocation of resources. This study places its focus on… Expand

8 PDF IOP Publishing Save Cite

Super Real-Time Forecast of Wildland Fire Spread by A Dual-Model Deep Learning Method

Y. Z. Li Z. L. Wang X. Y. Huang

Environmental Science, Computer Science · Journal of Environmental Informatics · 2024

TLDR The findings demonstrate the potential of AI in offering fast and high-resolution forecasts of wildfire spread, and the novel contribution

is to leverage two **models** which can **work** in **tandem and** be utilized at various stages of **wildfire management**. Expand

8 Publisher Save Cite

A high-fidelity ensemble simulation framework for interrogating wildland-fire behaviour and benchmarking machine learning models

Qing Wang Matthias Ihme Cenk Gazen Yi-Fan Chen John Anderson

Environmental Science, Computer Science · International journal of wildland fire · 12 June 2024

TLDR A high-fidelity ensemble **wildfire simulation** framework that integrates the Swirl-**Fire** large-eddy **simulation** tool **for wildfire predictions** with the Vizier optimisation platform **for** automated run-time **management** of ensemble **simulations and** large-scale batch processing is presented. Expand

2 [PDF] arXiv Save Cite

Predicting the rate of spread of mixed-fuel surface fires in northeastern China using the Rothermel wildfire behaviour model: a laboratory study

Hui Yang Huiying Cai Guang Yang Daotong Geng Long Sun

Environmental Science · Journal of Forest Research · 4 July 2024

The rate of **fire spread** is a key indicator for assessing **forest fire** risk and developing **fire management** plans. The Rothermel **model** is the most widely used **fire spread model**, established through… Expand

3 Springer Nature Save Cite

An Interactive Data-Driven HPC System for Forecasting Weather, Wildland Fire, and Smoke

J. Mandel M. Vejmelka +4 authors K. Hilburn

Environmental Science, Computer Science · 2019 IEEE/ACM HPC for Urgent Decision Making… · 1 November 2019

TLDR The **core** of the system is a coupled weather, **wildland fire**, fuel moisture, **and** smoke **model**, running in an interactive **workflow and** data **management** system that automates job setup, data acquisition, preprocessing, **and simulation** on an HPC cluster. Expand

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